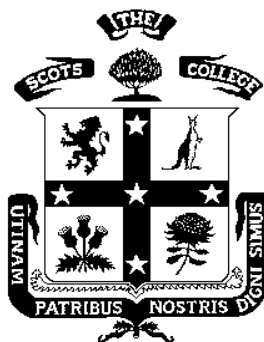

THE SCOTS COLLEGE



MATHEMATICS EXTENSION II

YEAR 12 PRETRIAL

23TH MARCH 2015

GENERAL INSTRUCTIONS

- Reading time – 5 minutes
- Working time - 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- Show all necessary working in Section II

WEIGHTING

30%

TOTAL MARKS

70

SECTION I (7 MARKS)

- Answers to be recorded on the multiple choice answer sheet provided
- Clearly label your answer sheet with your student number
- Allow about 10 minutes for this section

SECTION II (63 MARKS)

- Questions 8 - 11
- Answers to be recorded in the answer booklets provided
- *Each question must be completed in a new answer booklet.*
- Label each answer booklet with your student number and the question number attempted. Clearly indicate the booklet order if more than one booklet is used for a question. E.g. Book 1 of 2 and 2

Section 1

Question 1

The equation of a conic is $25x^2 - 16y^2 = -400$. The eccentricity of the conic is given by

A. $e = \frac{\sqrt{14}}{5}$

B. $e = \frac{\sqrt{41}}{15}$

C. $e = \frac{\sqrt{21}}{5}$

D. $e = \frac{\sqrt{41}}{5}$

Question 2

If $e^{x+y} = xy$ then $\frac{dy}{dx} =$

A. $\frac{y^{1-x}}{xy^{-1}}$

B. $\frac{y(x-1)}{x(y-1)}$

C. $\frac{y(1-x)}{x(y-1)}$

D. $\frac{x(1-y)}{y(x-1)}$

Question 3

If $z^2 = 4cis\left(\frac{4\pi}{3}\right)$ then $z =$

A. $1 - \sqrt{3}i$ or $-1 + \sqrt{3}i$

B. $3 + i$ or $-3 - i$

C. $-1 + \sqrt{3}i$ or $-1 - \sqrt{3}i$

D. $3 - i$ or $-3 + i$

Question 4

If $P(z) = z^3 - 2z^2 + 4z - 8$, and $z \in \mathbb{C}$ then a linear factor of $P(z)$ is

A. $z + 2i$

B. $z + 2$

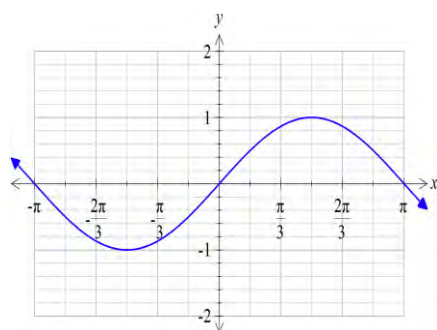
C. $2i$

D. 2

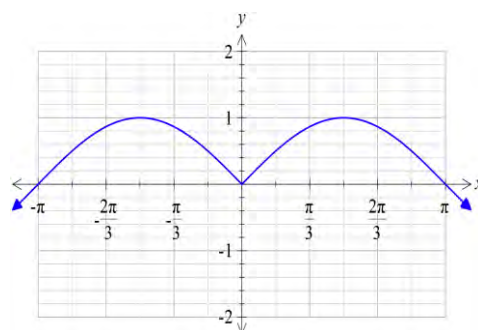
Question 5

Which one of the graphs below represents $|y| = |\sin x|$?

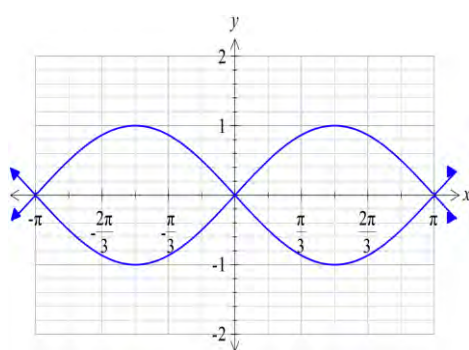
A.



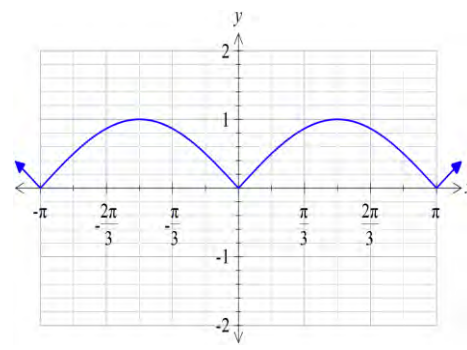
B.



C.



D.



Question 6

Let α, β, γ be the roots of the equation $x^3 + 4x^2 - 3x + 1 = 0$. The equation with roots $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$ is

A. $x^3 + 4x^2 - 3x + 1 = 0$

B. $x^3 - 3x^2 + 4x + 1 = 0$

C. $x^3 - 8x^2 + 12x + 3 = 0$

D. $x^3 - 4x^2 + 8x - 3 = 0$

Question 7

The line $y = mx + c$ will touch the hyperbola $xy = k$ if and only if

A. $c^2 - 4mk = 0$

B. $c^2 + 4mk = 0$

C. $k^2 + 4mc = 0$

D. $m^2 + 4ck = 0$

Section 2

Question 8 (Marks 13)

- a) Given the complex numbers $z_1 = \frac{p}{1+2i}$ and $z_2 = \frac{q}{1+i}$ where p and q are real, find p and q if $z_1 - z_2 = 4i$. [3]

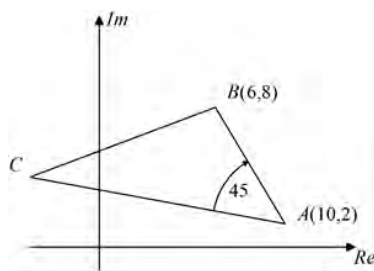
- b) Show that $z^n + z^{-n} = 2 \cos n\theta$ where $z = \cos \theta + i \sin \theta$. [4]
Hence or otherwise show that

$$\cos^4 \theta = \frac{\cos 4\theta + 4 \cos 2\theta + 3}{8}.$$

- c) The locus of a point (x, y) , which moves in the complex plane is represented by $|z - 3i| = 2$. [3]

- i) Sketch the locus on an Argand diagram.
ii) Show that the minimum value of $\arg z$ is $\cos^{-1}\left(\frac{2}{3}\right)$.
iii) Find the modulus of z when P is in the position of minimum argument.

- d) [3]



Triangle ABC is drawn on the Argand plane, where $\angle BAC = 45^\circ$, A represents the complex number $10 + 2i$ and B represents $6 + 8i$.

If the length of the side AC is twice the length of AB then find the complex number that point C represents.

Question 9 [marks 10]

(a) Consider the function $f(x) = \frac{e^x - 1}{e^x + 1}$. [7]

i) Show that the function is odd.

ii) Show that the function is always increasing.

iii) Find $f'(0)$.

iv) Sketch $f(x)$ showing any asymptotes.

v) Use your graph to find the values of k for which $\frac{e^x - 1}{e^x + 1} = kx$ has 3 real solutions.

(b) Given $f(x) = 1 - x^2$, without using any calculus, draw neat sketches of the following curves showing intercepts, asymptotes and turning points. The sketches should be about half a page each. [3]

i) $y = \frac{1}{f(x)}$

ii) $y = e^{f(x)}$

Question 10 [marks 21]

a) i) Show that the point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. [6]

ii) Find the equation of the tangent to above ellipse at P .

iii) Q is a point on the circle $x^2 + y^2 = a^2$ having the same x value as P . Write down the coordinates of Q .

iv) Show that the tangents at P and Q (provided $\theta \neq \frac{\pi}{2}$) meet on the x -axis.

- b) The hyperbola has equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$. [6]
- i) Sketch the hyperbola showing the coordinates of its foci, the equation of its directrices and asymptotes.
- ii) $P(4 \sec \theta, 3 \tan \theta)$, is a point on the hyperbola. Perpendiculars from P to the asymptotes meet these lines in M and N . Prove that $PM \cdot PN$ is independent of the position of P .
- c) The point $P\left(cp, \frac{c}{p}\right)$ lies on the rectangular hyperbola $xy = c^2$ in the first quadrant. The tangent to the hyperbola at the point P , crosses the x - axis at the point A and the y - axis at the point B . [9]
- i) Find the equation of the tangent to the hyperbola at the point P .
- ii) Show that the equation to the normal to the hyperbola at the point P is
- $$p^3x - py = cp^4 - c$$
- iii) If the normal at P meets the other branch of the hyperbola at the point Q , determine the coordinates of Q .
- iv) Show that the area of the triangle ABQ is
- $$c^2 \left(p^2 + \frac{1}{p^2}\right)^2$$
- v) Prove that the area of this triangle is a minimum when $p = 1$.
- Question 11 [marks 19]**
- a) For what values of r is $z - ri$ a factor of $P(z) = z^4 - z^3 + 9z^2 - 4z + 20$? [4]
- Hence or otherwise, solve $P(z) = 0$, $r \neq 0$ over the set of Complex numbers.
- b) Given that $P(x) = x^3 + 3px + q$ has a factor of $(x - k)^2$, [4]
- i) Show that $p = -k^2$
- ii) Find q in terms of k .
- iii) Hence verify that $4p^3 + q^2 = 0$.

- c) If α, β and γ are the roots of $x^3 - x^2 - 4x + 1 = 0$ find the equation whose roots are $\alpha + \beta - \gamma$, $\beta + \gamma - \alpha$, and $\gamma + \alpha - \beta$. [4]

Hence evaluate $(\alpha + \beta - \gamma)(\beta + \gamma - \alpha)(\gamma + \alpha - \beta)$.

- d) Given $\frac{3-x}{(1+6x)(1+2x)^2} = \frac{c_1}{(1+6x)} + \frac{c_2}{1+2x} + \frac{c_3}{(1+2x)^2}$, [3]

Find c_1 , c_2 and c_3 .

- e) Show that if the equation $x^n - ax^2 + b = 0$ has a multiple root, then [4]

$$n^n b^{n-2} = 4a^n(n-2)^{n-2}.$$

End of Exam

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



THE SCOTS COLLEGE-2015-MATHEMATICS EXTENSION 2
MATHEMATICS PRE-TRIAL HSC

CANDIDATE NUMBER: _____

SECTION I – MULTIPLE CHOICE ANSWER SHEET (7 MARKS)

Mark the correct answer by filling in the circle. To make a correction, neatly place a cross over the circle and then fill in the correct circle.

EXAMPLE:

A	B	C	D
☐	☒	☐	☐

	A	B	C	D
Question 1	☐	☐	☐	☐
Question 2	☐	☐	☐	☐
Question 3	☐	☐	☐	☐
Question 4	☐	☐	☐	☐
Question 5	☐	☐	☐	☐
Question 6	☐	☐	☐	☐
Question 7	☐	☐	☐	☐

①
Solutions ExT2

Multi Choice

Q8
(a) $z_1 = \frac{p}{1+2i}, z_2 = \frac{q}{1+i}$

$$z_1 = \frac{p}{1+2i} \times \frac{1-2i}{1-2i} \quad z_2 = \frac{q}{1+i} \times \frac{1-i}{1-i}$$

$$= \frac{p-2pi}{5} \quad = \frac{q-qi}{2}$$

$$z_1 - z_2 = 4i$$

$$\frac{p-2pi}{5} - \frac{q-qi}{2} = 4i$$

$$2p-4pi-5q+5qi=40i$$

$$2p-5q=0 \quad \text{--- (1) } \times 2$$

$$5q-4p=40 \quad \text{--- (2)}$$

use (1) in (2)

$$5q-10q=40$$

$$-5q=40$$

$$q=-8 \quad \checkmark$$

$$\therefore p = + \frac{5(-8)}{2}$$

$$= \frac{-40}{2}$$

$$= -20 \quad \checkmark$$

(b) Let $z = \cos \theta + i \sin \theta$

$$\therefore z^{-1} = \cos \theta - i \sin \theta$$

$$z + z^{-1} = 2 \cos \theta \quad \text{--- (1)}$$

$$\text{Now } z^2 + z^{-2} = (\cos \theta + i \sin \theta)^2 + (\cos \theta - i \sin \theta)^2$$

$$= \cos^2 \theta + 2i \sin \theta \cos \theta - \sin^2 \theta$$

$$+ \cos^2 \theta - 2i \sin \theta \cos \theta - \sin^2 \theta$$

$$= 2(\cos^2 \theta - \sin^2 \theta)$$

$$= 2 \cos 2\theta$$

Similarly

$$\therefore z^n + z^{-n} = 2 \cos n\theta$$

$$\text{From (1) } (z + z^{-1})^4 = 2^4 \cos^4 \theta$$

$$= 16 \cos^4 \theta$$

Expanding L.H.S.

$$z^4 + 4z^3z^{-1} + 6z^2z^{-2} + 4zz^{-3} + z^{-4} = 16 \cos^4 \theta$$

$$\Rightarrow (z^4 + z^{-4}) + 4z^2 + 6 + 4z^{-2} = 16 \cos^4 \theta$$

$$\Rightarrow 2 \cos 4\theta + 4(z^2 + z^{-2}) + 6 = 16 \cos^4 \theta$$

$$\Rightarrow 2 \cos 4\theta + 4 \times 2 \cos 2\theta + 6 = 16 \cos^4 \theta$$

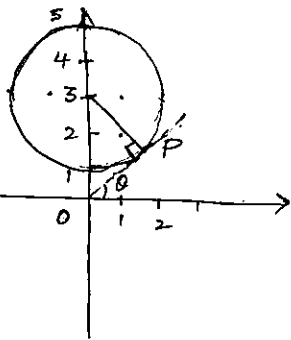
$$\Rightarrow \cos 4\theta + 4 \cos 2\theta + 3 = 8 \cos^4 \theta$$

$$\Rightarrow \cos 4\theta = \frac{\cos 4\theta + 4 \cos 2\theta + 3}{8}$$

- | | |
|---|---|
| 1 | D |
| 2 | C |
| 3 | C |
| 4 | A |
| 5 | C |
| 6 | B |
| 7 | B |

Q.8

(c). $|z - 3i| = 2$.



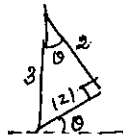
From diagram

$$\cos \theta = \frac{2}{3}$$

$$\theta = \cos^{-1}\left(\frac{2}{3}\right)$$

$$\text{i.e. } \arg z = \cos^{-1}\left(\frac{2}{3}\right)$$

$$|z| = \sqrt{3^2 - 2^2} = \sqrt{5}$$



$$(d) \vec{AB} = (6+8i) - (10+2i) \\ = -4+6i$$

$$\text{Now } \vec{OA} + \vec{AC} = \vec{OC}$$

$$\vec{AC} = 2 \cos 45^\circ \times \vec{AB}$$

$$= 2\left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right)(-4+6i)$$

$$= \sqrt{2}(1+i)(-4+6i)$$

$$= \sqrt{2}(-10+2i)$$

$$\therefore \vec{OC} = (10+2i) + \sqrt{2}(-10+2i)$$

$$= 10(1-\sqrt{2}) + 2i(1+\sqrt{2}) \checkmark$$

Q9.

(3)

$$a) i) f(x) = \frac{e^x - 1}{e^x + 1}$$

$$f(-x) = \frac{e^{-x} - 1}{e^{-x} + 1}$$

$$= \frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 1}$$

$$= \frac{1 - e^x}{1 + e^x}$$

$$= \frac{1 - e^x}{1 + e^x}$$

$$= - \frac{e^x - 1}{e^x + 1}$$

$$= -f(x)$$

∴ Function is odd.

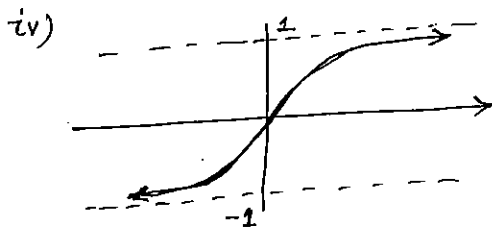
$$ii) f'(x) = \frac{(e^x + 1)e^x - (e^x - 1)e^x}{(e^x + 1)^2}$$

$$= \frac{2e^x}{(e^x + 1)^2}$$

> 0 for all x, $e^x > 0$

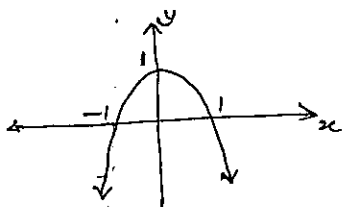
∴ function is an increasing function.

$$iii) f'(0) = \frac{2e^0}{(e^0 + 1)^2} = \frac{1}{2}$$

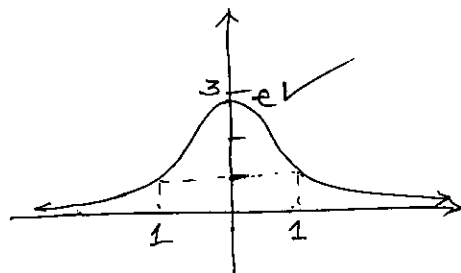


$$v) 0 < k < 1/2$$

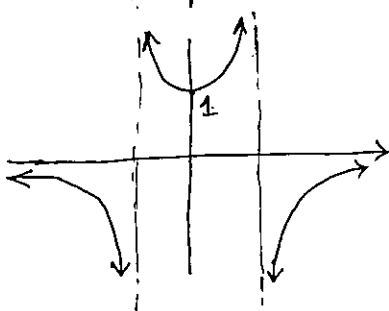
$$b) f(x) = 1 - x^2$$



(ii)



(i)



asymptotes ✓
graph ✓

(4)

Q10

$$a) (i) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\begin{aligned} \text{L.H.S.} &= \frac{a^2 \cos^2 \theta}{a^2} + \frac{b^2 \sin^2 \theta}{b^2} \\ &= \cos^2 \theta + \sin^2 \theta \\ &= 1 = \text{R.H.S.} \end{aligned}$$

$\therefore P$ lies on the ellipse

ii)

$$\text{At } P, x = a \cos \theta, y = b \sin \theta$$

$$\frac{dx}{d\theta} = -a \sin \theta \quad \frac{dy}{d\theta} = b \cos \theta$$

$$\frac{dy}{dx} = -\frac{b \cos \theta}{a \sin \theta}$$

$\therefore$ tangent at P is:

$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a \sin \theta y - a b \sin^2 \theta = -b \cos \theta x + a b \cos^2 \theta$$

$$a \sin \theta y + b \cos \theta x = a b (\sin^2 \theta + \cos^2 \theta)$$

$$a \sin \theta y + b \cos \theta x = a b \quad \checkmark \leftarrow (A)$$

$$iii) Q = (a \cos \theta, a \sin \theta)$$

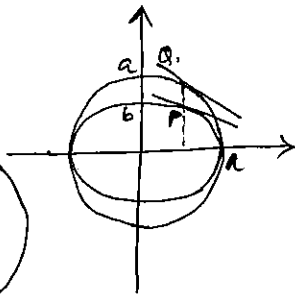
$$\therefore x^2 + y^2 = a^2$$

$$a^2 \cos^2 \theta + y^2 = a^2$$

$$y^2 = a^2 (1 - \cos^2 \theta)$$

$$y^2 = a^2 \sin^2 \theta$$

$$y = a \sin \theta$$



iv) at Q .

$$x = a \cos \theta \quad y = a \sin \theta$$

$$\frac{dx}{d\theta} = -a \sin \theta \quad \frac{dy}{d\theta} = a \cos \theta$$

$$\frac{dy}{dx} = -\frac{\cos \theta}{\sin \theta}$$

$\therefore$ tangent at Q is:

$$y - a \sin \theta = -\frac{\cos \theta}{\sin \theta} (x - a \cos \theta)$$

$$\sin \theta y - a \sin^2 \theta = -\cos \theta x + a \cos^2 \theta$$

$$\cos \theta x + \sin \theta y = a (\sin^2 \theta + \cos^2 \theta)$$

$$\cos \theta x + \sin \theta y = a \quad \checkmark \leftarrow (B)$$

$$y = 0 \text{ in } (A) \Rightarrow x = \frac{a}{\cos \theta}$$

$$y = 0 \text{ in } (B) \Rightarrow x = \frac{a}{\cos \theta}$$

Both tangents have the same x -intercept, hence they meet on x -axis.

5

Q10 (b)

$$i) \frac{x^2}{16} - \frac{y^2}{9} = 1$$

$$\frac{a}{16} = e^2 - 1$$

$$e^2 = \frac{25}{16}$$

$$e = \frac{5}{4} \quad \checkmark$$

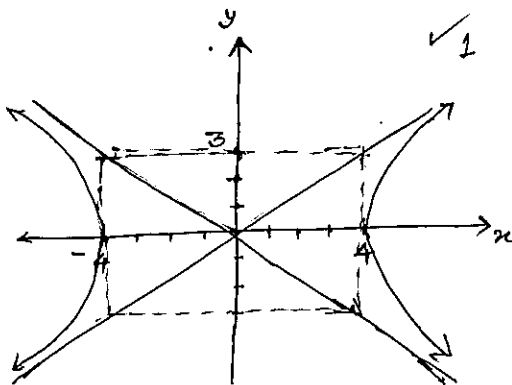
$$S(\pm ae, 0) = (\pm 5, 0)$$

directrices

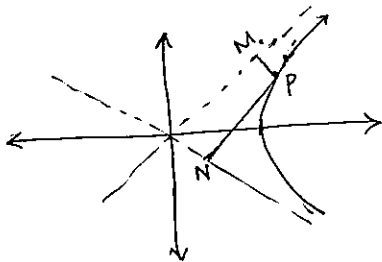
$$x = \pm \frac{a}{e} = \pm \frac{16}{5} \\ = \pm 3\frac{1}{5} \quad \checkmark$$

asymptotes.

$$y = \pm \frac{b}{a} x \\ = \pm \frac{3}{4} x. \quad \checkmark$$



ii)



$$P(4\sec\theta, 3\tan\theta)$$

asymptotes

$$3x + 4y = 0$$

$$3x - 4y = 0$$

$$PM = \frac{|12\sec\theta - 12\tan\theta|}{\sqrt{3^2 + 4^2}} \\ = \frac{12|\sec\theta - \tan\theta|}{5}$$

$$PN = \frac{12|\sec\theta + \tan\theta|}{5}$$

$$PM \cdot PN = \frac{144 |\sec^2\theta - \tan^2\theta|}{25}$$

$$= \frac{144}{25}$$

This is a constant which is independent of θ

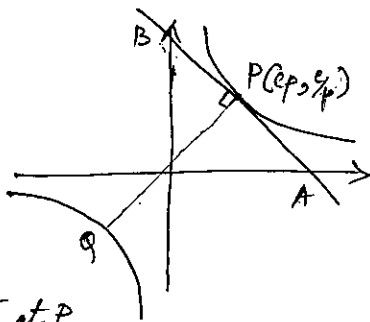
(6)

Q10 c)

$$i) xy = c^2$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

$$\text{at } P, \frac{dy}{dx} = -\frac{1}{p^2}$$



∴ Equation of tangent at P

$$y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$$

$$x + p^2y = 2cp \quad \checkmark$$

(ii) Equation of Normal

$$y - \frac{c}{p} = p^2(x - cp)$$

$$py - c = p^3x - cp^4$$

$$p^3x - py = cp^4 - c \quad \checkmark \text{--- (A)}$$

$$(iii) xy = c^2$$

$$y = \frac{c^2}{x} \text{ sub in (A)}$$

$$p^3x - p\frac{c^2}{x} = cp^4 - c$$

$$p^3x^2 - pc^2 = cp^4x - cx$$

$$p^3x^2 - (cp^4 - c)x - c^2p = 0$$

$$(x - cp)(p^3x + c) = 0$$

$$\Rightarrow x = -\frac{c}{p^3} \text{ at } Q \text{ and } y = -cp^3$$

$$\therefore Q\left(-\frac{c}{p^3}, -cp^3\right)$$

$$(iv) A(2cp, 0) \quad B(0, \frac{2c}{p})$$

$$d_{AB} = \sqrt{4c^2p^2 + \frac{4c^2}{p^2}} = \frac{2c}{p}\sqrt{p^4 + 1}$$

$$d_{PQ} = \sqrt{\left(cp + \frac{c}{p^3}\right)^2 + \left(\frac{c}{p} + cp^3\right)^2}$$

$$= \frac{cp^4 + c}{p^3} \sqrt{1 + p^4}$$

$$\text{Area of } \triangle = \frac{1}{2} AB \times PQ$$

$$= \frac{1}{2} \cdot \frac{2c}{p} \sqrt{p^4 + 1} \times \frac{cp^4 + c}{p^3} \sqrt{1 + p^4}$$

$$= \frac{c^2}{p^4} (p^4 + 1)^2$$

$$= c^2 \left(p^2 + \frac{1}{p^2}\right)^2 \quad \checkmark$$

$$(v) A = c^2 \left(p^2 + \frac{1}{p^2}\right)^2$$

$$A = c^2 \left(p^4 + 2 + \frac{1}{p^4}\right)$$

$$\frac{dA}{dp} = c^2 \left(4p^3 - \frac{4}{p^5}\right)$$

$$\frac{dA}{dp} = 0 \Rightarrow p^3 = \frac{1}{p^5}$$

$$= p^8 = 1$$

$$p = 1$$

$$\frac{d^2A}{dp^2} = 4c^2 (3p^2 + 5p^{-6}) > 0$$

∴ Area is minimum.

7

Q11 a)

$$(i) \quad p(z) = z^4 - z^3 + 9z^2 - 4z + 20$$

$$p(ri) = (ri)^4 - (ri)^3 + 9(ri)^2 - 4(ri) + 20 = 0$$

$$\Rightarrow r^4 + r^3i - 9r^2 - 4ri + 20 = 0$$

$$\Rightarrow r^4 - 9r^2 + 20 + i(r^3 - 4r) = 0$$

$$r^4 - 9r^2 + 20 = 0 \quad \text{and} \quad r^3 - 4r = 0$$

$$(r^2 - 5)(r^2 - 4) = 0 \quad r(r^2 - 4) = 0$$

$$r^2 = 5 \quad \text{or} \quad r^2 = 4 \quad r \neq 0 \quad \text{or} \quad r^2 = 4$$

$$r = \pm 2$$

$$r = \pm 2$$

✓
($r^2 = 5$ satisfies only one solution)

∴ two roots are $2i, -2i$

$$\text{Now } (z - 2i)(z + 2i) = z^2 + 4$$

$$\begin{array}{r} z^2 + 4 \overline{) z^4 - z^3 + 9z^2 - 4z + 20} \\ \underline{-z^4 \quad +4z^2} \\ -z^3 + 5z^2 - 4z \\ \underline{+z^3 \quad -4z} \\ 5z^2 + 20 \\ \underline{-5z^2} \\ 20 \end{array}$$

$$\therefore p(z) = (z^2 + 4)(z^2 - z + 5)$$

$$z^2 - z + 5 = 0$$

$$\Rightarrow z = \frac{1 \pm \sqrt{1^2 - 4 \times 5}}{2}$$

$$= \frac{1 \pm \sqrt{-19}}{2}$$

$$= \frac{1 \pm i\sqrt{19}}{2}$$

(8)

Q 11 b)

$$i) P(x) = x^3 + 3px + q$$

has a factor $(x-k)^2$

$$\text{then } P'(x) = 3x^2 + 3p$$

has a factor $(x-k)$ ✓

$$\therefore P'(k) = 3k^2 + 3p = 0$$

$$3p = -3k^2$$

$$p = -k^2 \quad \checkmark$$

$$ii) P(k) = k^3 + 3pk + q = 0$$

$$\text{sub } p = -k^2$$

$$k^3 + 3(-k^2)k + q = 0$$

$$k^3 - 3k^3 + q = 0$$

$$\therefore q = 2k^3 \quad \checkmark$$

$$iii) \text{ show } 4p^3 + q^2 = 0.$$

sub for p and q

$$L.H.S = 4(-k^2)^3 + (2k^3)^2$$

$$= -4k^6 + 4k^6$$

$$= 0.$$

$$= R.H.S. \quad \checkmark$$

$$(c) \quad x^3 - x^2 - 4x + 1 = 0.$$

$$\text{we know } \alpha + \beta + \gamma = 1$$

$$\Rightarrow \alpha + \beta + 2\gamma - \gamma = 1$$

$$\alpha + \beta - \gamma = 1 - 2\gamma$$

$$\text{let } y = 1 - 2x.$$

$$\therefore x = \frac{1-y}{2}.$$

the equation is

$$\left(\frac{1-y}{2}\right)^3 - \left(\frac{1-y}{2}\right)^2 - 4\left(\frac{1-y}{2}\right) + 1 = 0. \quad \checkmark$$

$$\Rightarrow y^3 - y^2 - 17y + 9 = 0.$$

$$\text{OR } \Rightarrow x^3 - x^2 - 17x + 9 = 0 \quad \checkmark$$

Prod. of roots.

$$(\alpha + \beta - \gamma)(\beta + \gamma - \alpha)(\gamma + \alpha - \beta) = -9 \quad \checkmark$$

Q11 d)

(9)

$$\frac{3-x}{(1+6x)(1+2x)^2} = \frac{c_1}{1+6x} + \frac{c_2}{1+2x} + \frac{c_3}{(1+2x)^2}$$

$$3-x = c_1(1+2x)^2 + c_2(1+6x)(1+2x) + c_3(1+6x)$$

$$x = -\frac{1}{6} \Rightarrow 3 + \frac{1}{6} = c_1(1 + 2 \times \frac{1}{6})^2$$

$$\frac{19}{6} = c_1(\frac{4}{6})^2$$

$$c_1 = \frac{19 \times 6}{4^2} = \frac{57}{8} \checkmark$$

$$x = -\frac{1}{2} \Rightarrow 3 + \frac{1}{2} = c_3(1 + 6 \times -\frac{1}{2})$$

$$\frac{7}{2} = c_3(-2)$$

$$-\frac{7}{4} = c_3 \checkmark$$

$$\text{Coeff. of } x^2 \Rightarrow 0 = 4c_1 + 12c_2$$

$$0 = 4 \times \frac{57}{8} + 12c_2$$

$$12c_2 = -\frac{57}{2}$$

$$c_2 = -\frac{57}{12 \times 2} = -\frac{19}{8} \checkmark$$

$$(e) P(x) = x^n - ax^2 + b$$

$$P'(x) = nx^{n-1} - 2ax$$

Let x be a multiple root

$$x^n - ax^2 + b = 0 \quad \text{--- (1)}$$

$$nx^{n-1} - 2ax = 0 \quad \text{--- (2)}$$

$$nx \text{ (1)} \Rightarrow nx^n - anx^2 + bn = 0 \quad \text{--- (3)}$$

$$x \text{ (2)} \Rightarrow nx^n - 2ax^2 = 0 \quad \text{--- (4)}$$

$$(4) - (3) \Rightarrow ax^2(-2+n) - bn = 0$$

$$x^2 = \frac{bn}{a(n-2)}$$

$$\text{Sub in (4)} \quad n \left[\frac{bn}{a(n-2)} \right]^{n/2} - 2a \left[\frac{bn}{a(n-2)} \right] = 0$$

$$\Rightarrow n \left[\frac{bn}{a(n-2)} \right]^{n/2} = 2a \left[\frac{bn}{a(n-2)} \right]$$

$$\text{squaring} \quad \frac{n^2 b^n n^n}{a^n (n-2)^n} = \frac{4a^2 b^2 n^2}{a^2 (n-2)^2}$$

$$n^n b^{n-2} = 4a^n (n-2)^{n-2} \checkmark$$